IBM Z DATATHON 2026 · TEAM ZEUS CODE

AtmosIQ

Where Weather Meets Intelligence

Weather Analytics, Data Quality and Machine Learning

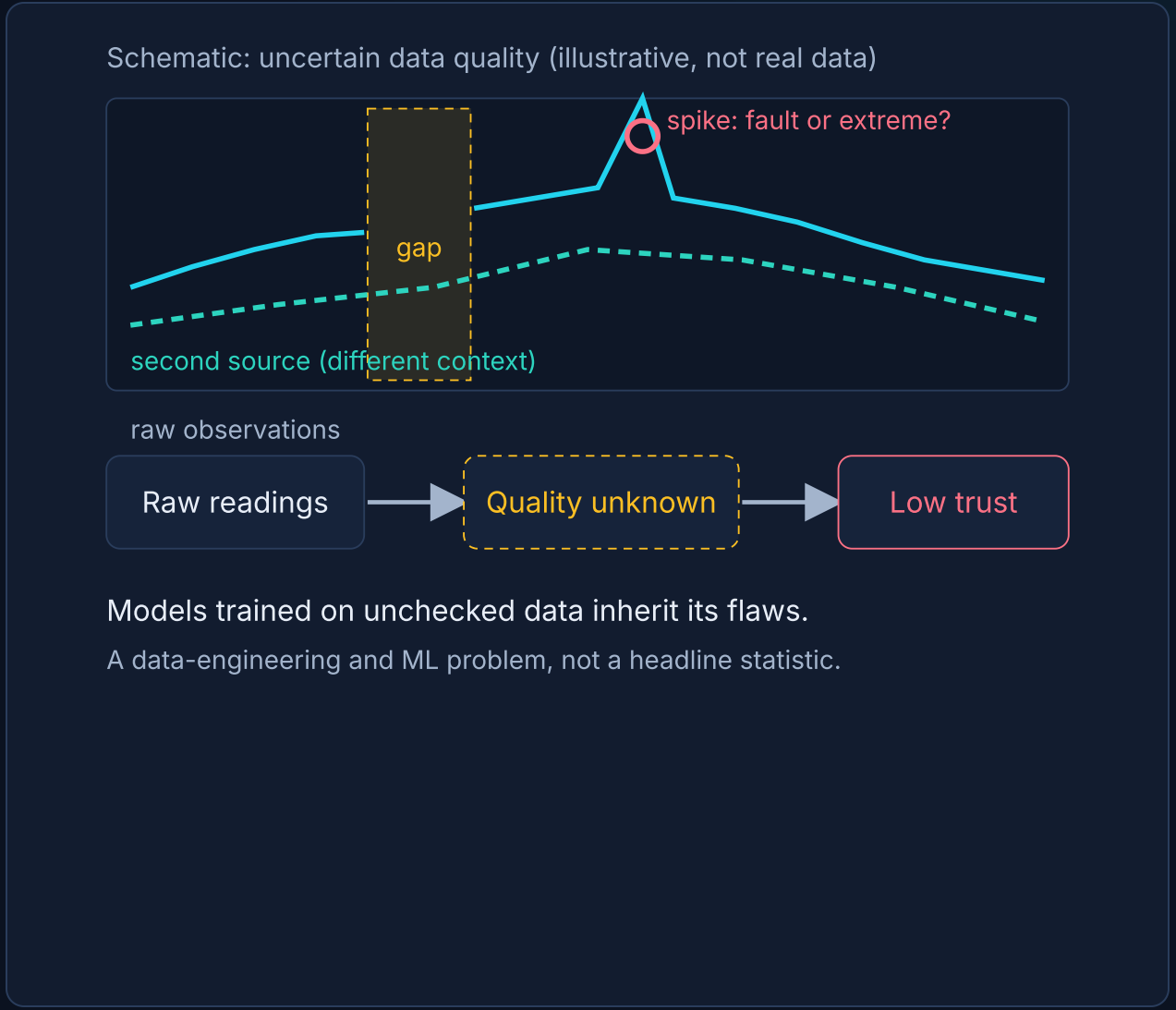
ZEUS CODE | IBM Z DATATHON 2026 | GERMANY WEATHER DATA

Python · FastAPI · scikit-learn

THE PROBLEM

Weather data is easy to collect and hard to trust

- **Missing readings** and incomplete timestamps leave gaps in hourly series.
- **Suspicious spikes** may be sensor faults, or real extremes.
- **Source differences:** station observations and gridded estimates describe different measurement contexts.
- **Convincing forecasts** can mislead without independent evaluation.
- Quality checks, history, predictions and evaluation usually live in **separate workflows**.



Models trained on unchecked data inherit its flaws.

A data-engineering and ML problem, not a headline statistic.

OUR SOLUTION

AtmosIQ: explore, monitor, predict, evaluate — in one platform

01
Explore

- Historical trends and daily summaries
- Interactive maps
- City-level conditions

02
Monitor

- Rule-based checks
- Time-segment analysis
- Isolation Forest anomaly detection

03
Predict

- Next-day daily min / max temperature
- Next-day daily min / max relative humidity
- Four independent regression models

04
Evaluate

- MAE, RMSE, R^2, tolerance-hit rates
- Data-source information
- Exportable evaluation reports

Read flags carefully. An anomaly flag marks an observation to investigate, not necessarily an incorrect measurement.

Also included: illustrative city-level threshold alerts (not official weather warnings) and an optional seasonal outlook from an external provider.

DATA FOUNDATION

Three weather datasets, three distinct roles

DATASET	FILE	NATURE	USED FOR
Meteostat-derived hourly data	`data/meteostat_2024.csv`	Hourly series	Default next-day forecasting pipeline. 8,784 rows; 432 missing temperature readings.
ECMWF IFS gridded archive	`data/reanalysis_hourly_2024_2026.csv`	Gridded estimates	Historical exploration and a separate city-level forecast layer. Nine sample cities.
German Weather Service (DWD) station archive	`data/dwd_station_observations_2024_2026.csv`	Station observations	City conditions, station summaries, comparisons, illustrative threshold alerts. Through 16 Oct 2026; 2026 values provisional.

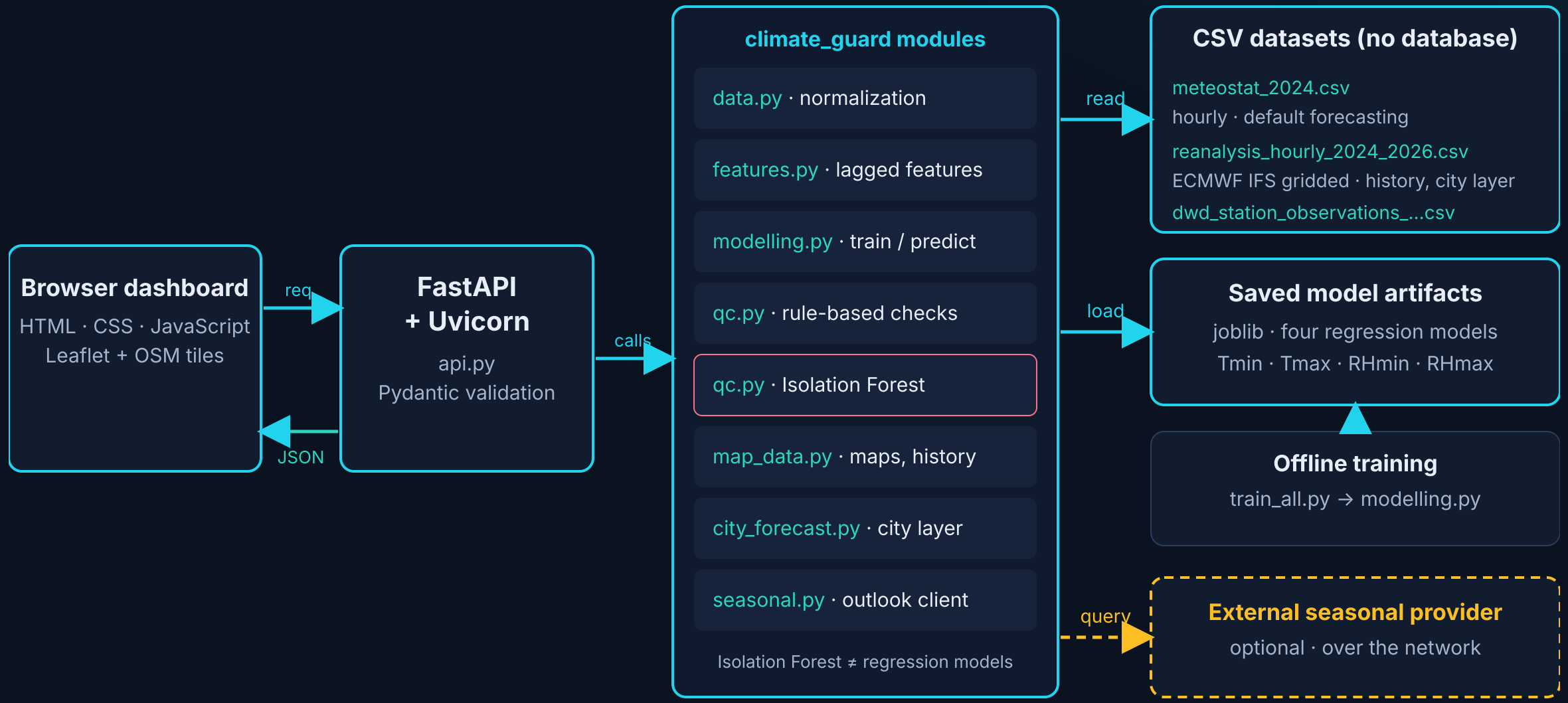
Not one merged dataset. The three files serve different purposes and are not automatically combined into a single training set.

Storage: CSV files plus saved model artifacts. No database engine is included in the project.

Source: project README (supplied documentation)

SYSTEM ARCHITECTURE

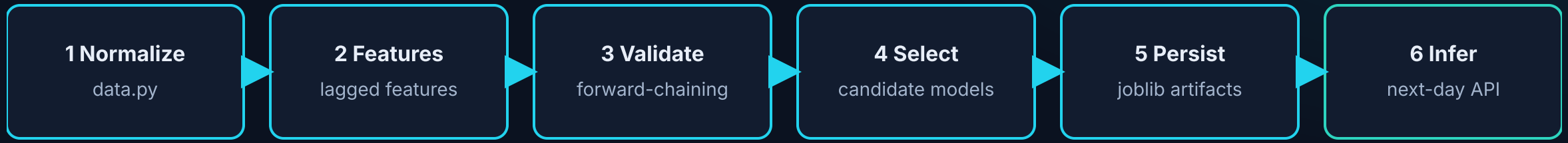
Implemented application architecture



Solid cyan = implemented · dashed amber = optional external dependency

MACHINE-LEARNING PIPELINE

From hourly readings to four next-day predictions



FOUR INDEPENDENT REGRESSION TARGETS

- Daily min temperature
- Daily max temperature
- Daily min rel. humidity
- Daily max rel. humidity

Candidates: persistence baselines, Ridge, Random Forest, Extra Trees, HistGradientBoosting. Trained from `meteostat_2024.csv` by default.

SEPARATE QUALITY-MONITORING MODEL

Isolation Forest

- Flags unusual observations alongside rule-based checks
- Does not predict weather; not one of the four regressors

DATA-QUALITY MONITORING

Three layers of checks produce flags to investigate

Rule-based checks

Explicit conditions on readings and timestamps, e.g. missing or incomplete data.

Time-segment analysis

Examines how data behaves across periods, so problems are located in time.

Isolation Forest

Model-based detection of unusual observations that simple rules may miss.

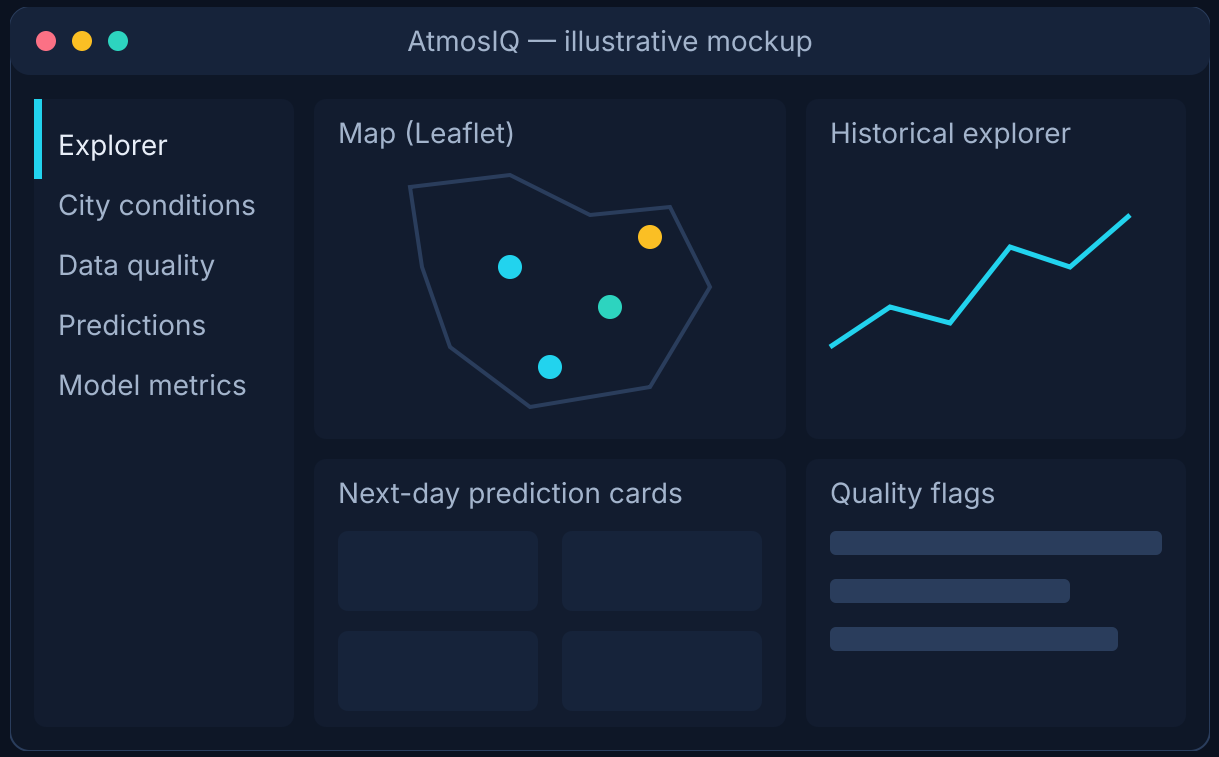


An anomaly flag is a prompt to investigate, not proof of a bad reading.

A real weather extreme can be flagged too. Findings carry supporting evidence so an analyst can decide.

DASHBOARD AND API

One browser interface over a small, validated JSON API



Illustrative mockup — not a screenshot of the running application. Placeholder shapes carry no data.

REQUEST WORKFLOW

1 · User action in the browser sends an HTTP request

2 · FastAPI validates it with Pydantic

3 · Processing reads a CSV or loads a saved model

4 · Prediction, quality analysis, aggregation or alert

5 · JSON response rendered in the dashboard

ENDPOINT GROUPS

history & maps · city conditions · data quality · next-day prediction · evaluation export · seasonal outlook

Endpoint groups summarised from project documentation; exact paths not shown

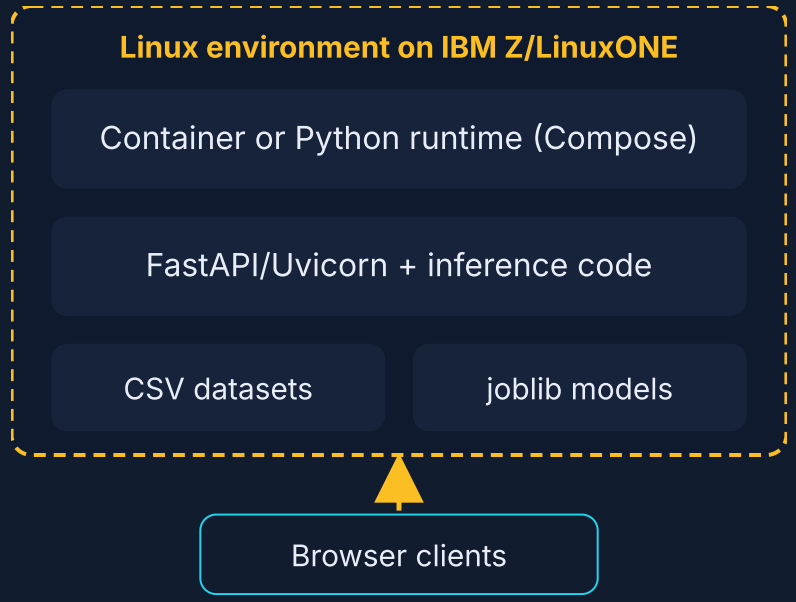
IBM Z/LinuxONE Deployment Architecture

PROPOSED

IMPLEMENTED

- Python 3.13 FastAPI/Uvicorn app and dashboard
- CSV datasets and joblib model artifacts
- Docker Compose, optional deployment
- pytest and HTTPX tests

PROPOSED HOSTING



Not deployed or tested on IBM Z as far as the supplied materials show.

Why evaluate here

A standard Python API with file-based data and saved-model inference is a Linux workload. Running it on LinuxONE would test portability and operational fit. No benefit is claimed or measured.

Unverified

- Actual deployment on IBM Z
- Distribution, instance specs
- Performance or cost figures
- Whether training or inference ran on Z

VALIDATION AND TECHNICAL CREDIBILITY

What we measure, and what the results do not prove

Metrics reported in evaluation metadata

- **MAE, RMSE**: typical and large-error size
- **R^2**: variance explained
- **Tolerance-hit rate**: share of predictions within a set tolerance. Not classification accuracy.

Leakage prevention

Chronological, forward-chaining validation keeps future data out of training folds.



Exploratory, not independent

Retrospective holdout metrics were examined during earlier model iterations. They are not proof of future performance, so no scores are shown. Independent future-period evaluation is still needed.

Tests and reproducibility

- pytest + HTTPX: 13 passed, 1 failed in the reported run
- Failure: exporter test expected 282 holdout-detail rows, got 7,308. A known documented mismatch to resolve.
- Pinned scikit-learn 1.8.0, `train_all.py`, joblib artifacts, Docker Compose

Source: project README test report

LIMITATIONS AND ROADMAP

Documented limits today, proposed improvements next

CURRENT LIMITATIONS (DOCUMENTED)

- No built-in authentication or authorization
- Bundled Meteostat-derived data is historical 2024, not a live feed
- Next-day endpoint needs enough recent history and a complete latest day
- Holdout results are exploratory
- One failing exporter test
- CSV storage; DWD 2026 values are provisional

ROADMAP (PROPOSED, NOT IMPLEMENTED)

- Independent future-period evaluation
- Clearer data provenance across the three sources
- Stronger API security (authentication, authorization)
- Expanded tests, including the exporter fix
- Observability: logging and metrics
- Deployment validation on IBM Z/LinuxONE

CONCLUSION AND RESOURCES

Weather data you can explore, question and verify

AtmosIQ joins exploration, data-quality monitoring, four next-day regressors and transparent evaluation in one Python platform, with its limits stated openly.

Zeus Code · IBM Z Datathon 2026

Repository URL: **to be confirmed**
Project software license: **to be confirmed**
IBM Z deployment evidence: **none supplied**

RESOURCE	TERMS / ATTRIBUTION
FastAPI, scikit-learn, pandas, NumPy, Leaflet fastapi.tiangolo.com · scikit-learn.org/stable · pandas.pydata.org/docs · numpy.org/doc · leafletjs.com	Open-source software; each project has its own license. Confirm before redistribution.
OpenStreetMap tiles openstreetmap.org/copyright	Map data attribution required per OSM copyright page.
Meteostat dev.meteostat.net/license	Data page states CC BY 4.0: credit Meteostat and original providers.
ECMWF open data ecmwf.int/en/forecasts/datasets/open-data	Open IFS data: CC-BY-4.0 plus ECMWF Terms of Use. Confirm our archive's exact origin.
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